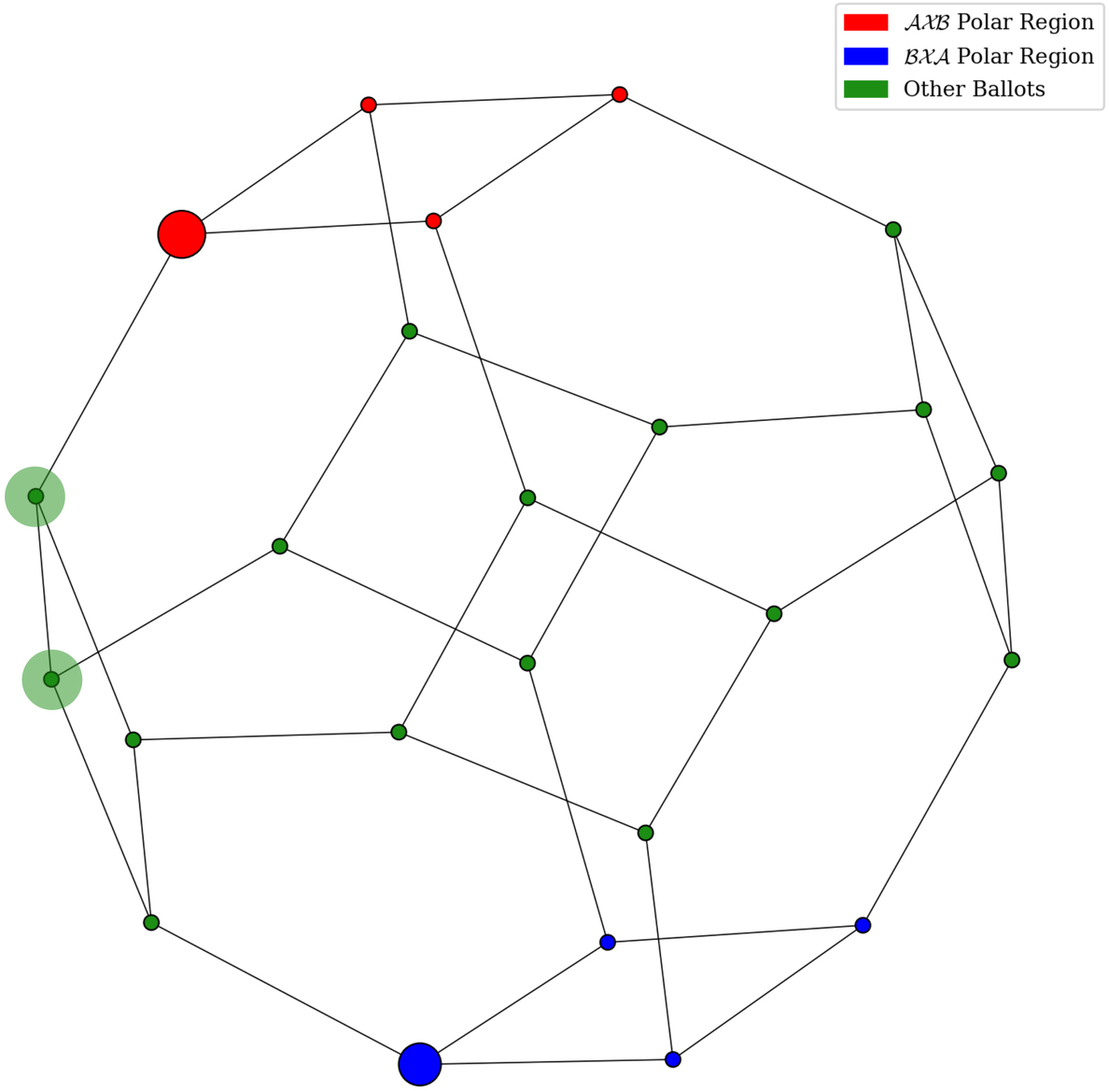
Measuring Polarization in Ranked Choice Voting Elections with Wasserstein Distance

By **polarized** there are (two) distinct groups that vote one slate of candidates over the other slate of candidates.

Example:

$$\begin{array}{cccc} 50 & 5 & 5 & 40 \\ \begin{pmatrix} A_1 \\ A_2 \\ B_1 \\ B_2 \end{pmatrix} & \begin{pmatrix} A_1 \\ B_1 \\ A_2 \\ B_2 \end{pmatrix} & \begin{pmatrix} A_1 \\ B_1 \\ B_2 \\ A_2 \end{pmatrix} & \begin{pmatrix} B_1 \\ B_2 \\ A_1 \\ A_2 \end{pmatrix} \end{array}$$

Construct a ballot graph $\mathscr{G}_m$ whose vertices are the ballots and there is an edge between two ballots if they differ by one neighbor swap.



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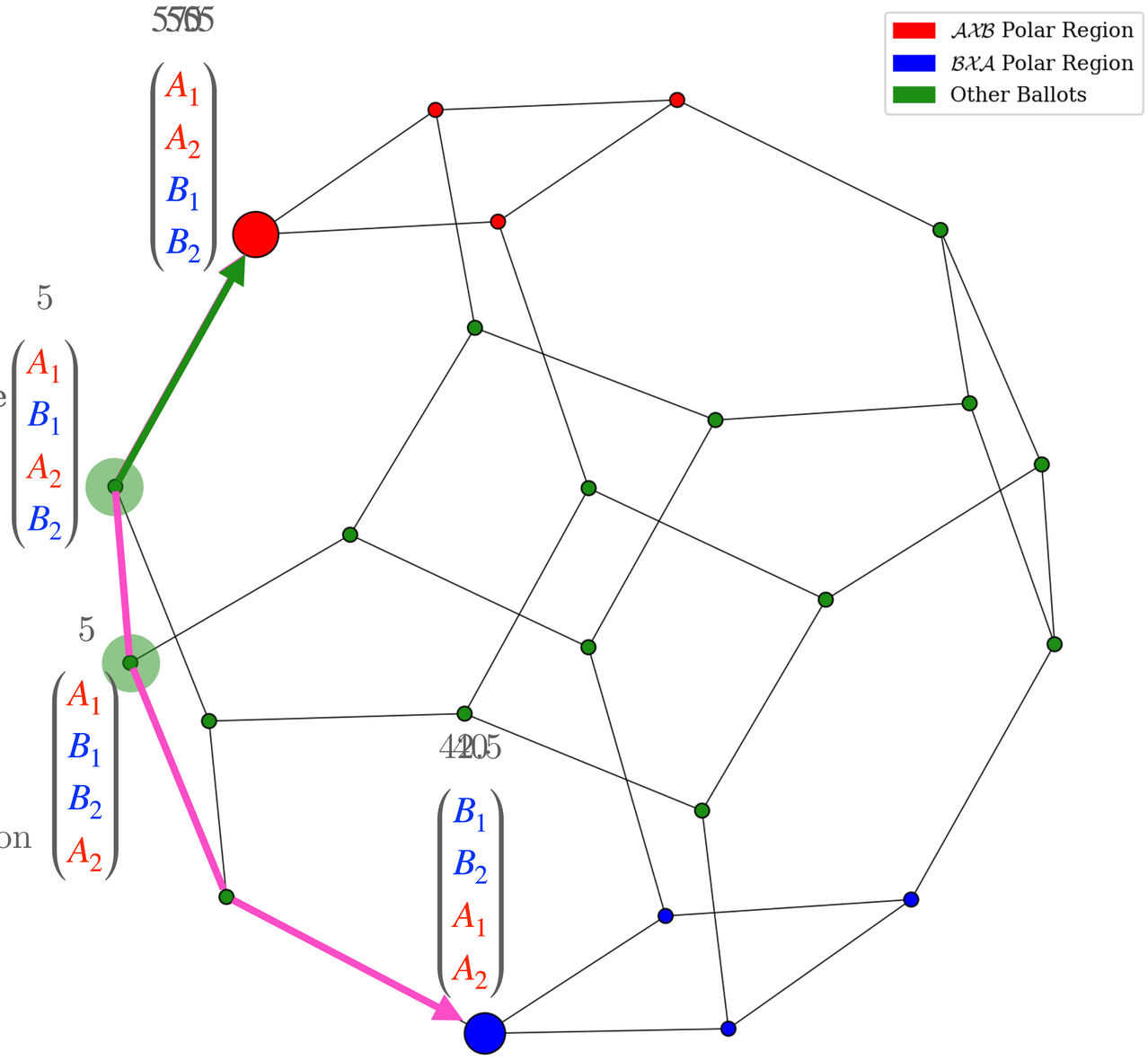
To measure how polarized this election is, we can consider the cumulative work needed to push the ballots to the ballot to the nearest polar region.

Cumulative work so far: 15

The cumulative work done is also the **1-Wasserstein distance** between the election (initial distribution) and the push forward polarized election (target distribution).

Smaller distance $\implies$ Closer to polarized distribution (more polarized)

Larger distance $\implies$ Farther away from polarized distribution (less polarized)



Many things to consider!

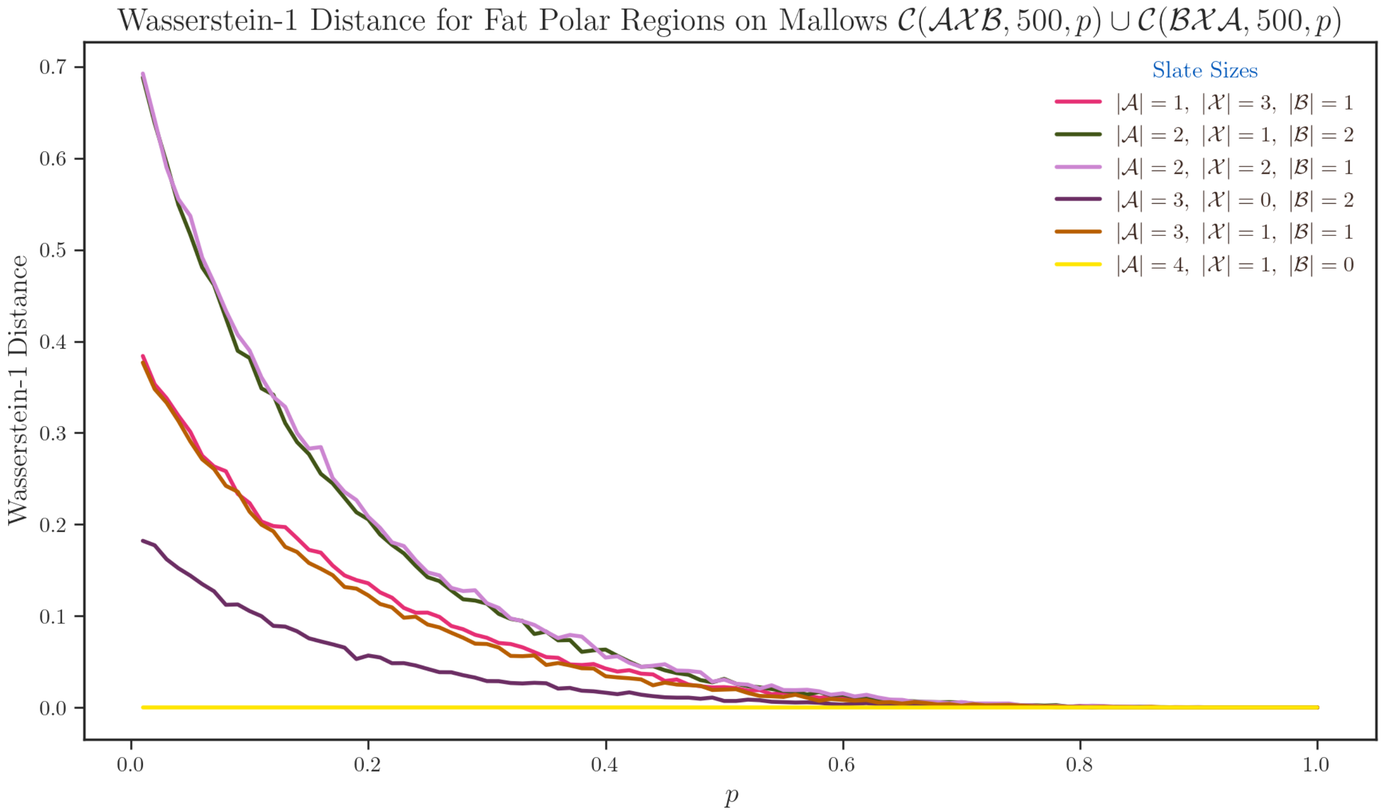
For now, validating with synthetic clustered elections

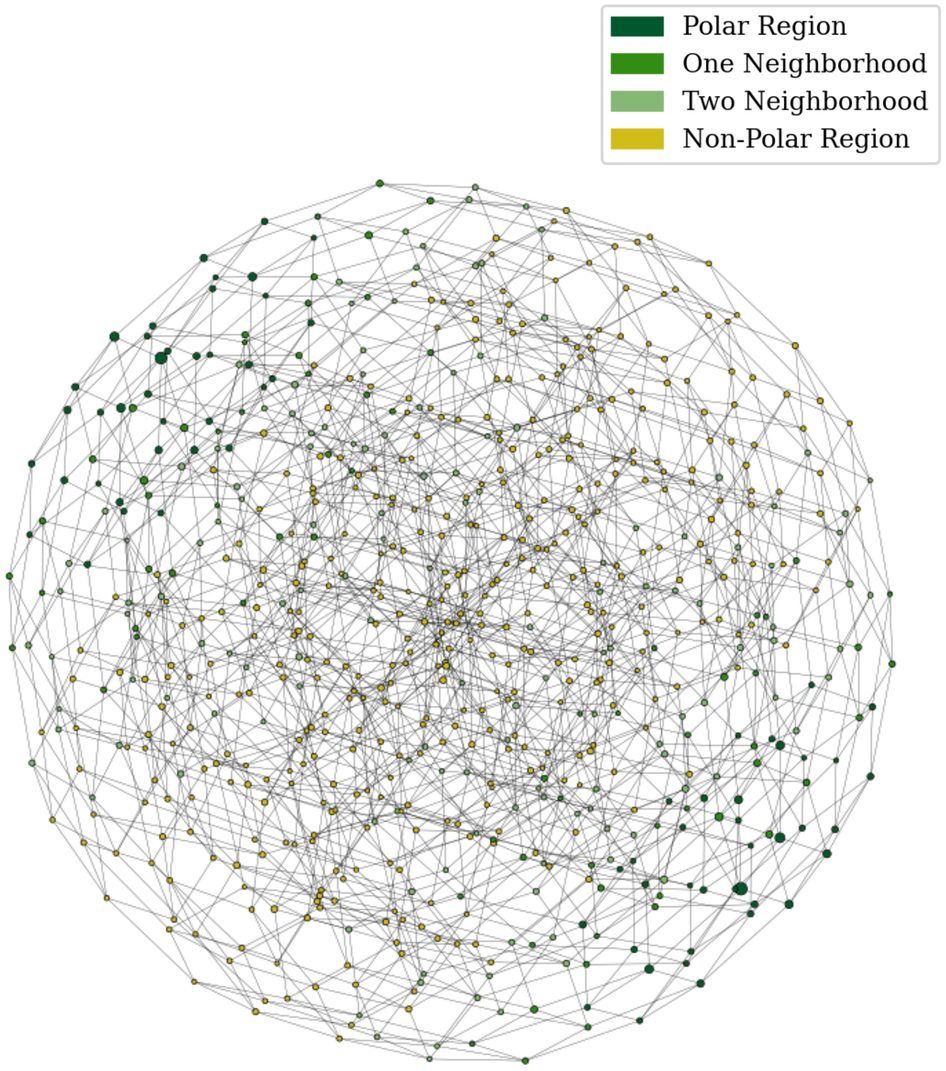
Mixed Mallow Model with clusters $\mathscr{C}(\mathscr{A}\mathscr{X}\mathscr{B}, 500, p) \cup \mathscr{C}(\mathscr{B}\mathscr{X}\mathscr{A}, 500, p)$

Generate a clustered election by

1. Select a center ballot. For example: $(A_1, A_2, X_1, B_1, B_2, B_3)$
2. Flip a weighted p weighted coins for tails and count the number of failures until tails (draw $X \sim \text{Geo}(p)$).
3. Do random swaps based on the number of failures until tails. This is a new ballot.

What does the 1-Wasserstein distance look like on these elections?





$p = 0.01$

Thanks!